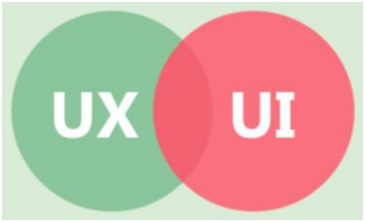
UX/UI for Mobile : Ergonomic Principles, Material You & Accessibility



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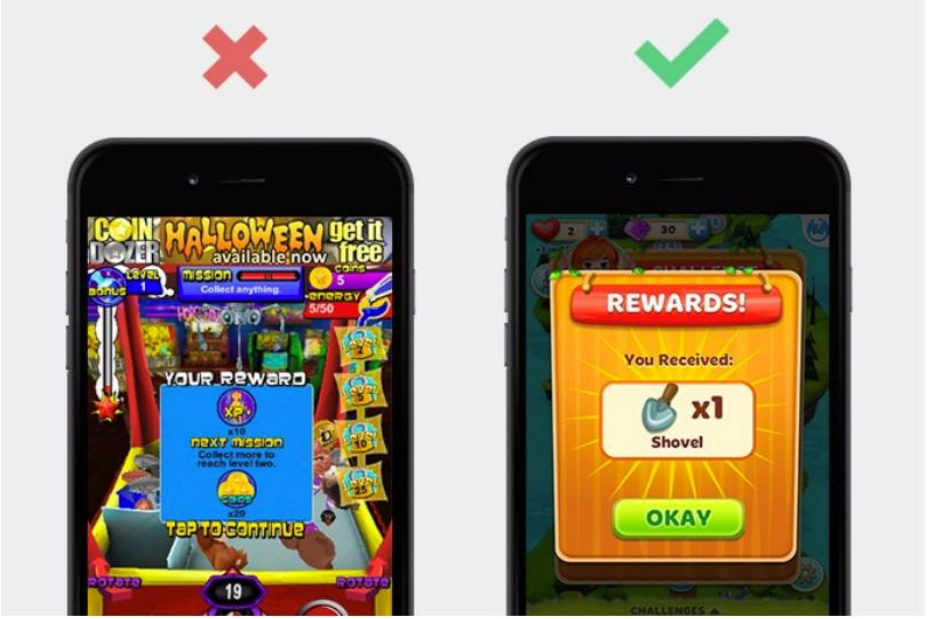
II.3510 – Mobile Application Development for Android

Why UX/UI Matters in Mobile Development

What frustrates you most when using a mobile app?

- Mobile apps are used everywhere, every time
- 90% of app uninstalls are due to bad UX or unclear UI
- Small screens and touch navigation = new ergonomic challenges
- Good UX -> more engagement, loyalty, and positive reviews
- Poor UX -> frustration, bad ratings, and uninstalls



What is UX ? What is UI ?

UI

- “User Interface” = how the app looks
- Consistent color palette and typography
- Proper spacing and alignment for readability
- Interactive elements that look “clickable”
- Visual hierarchy (important things stand out)

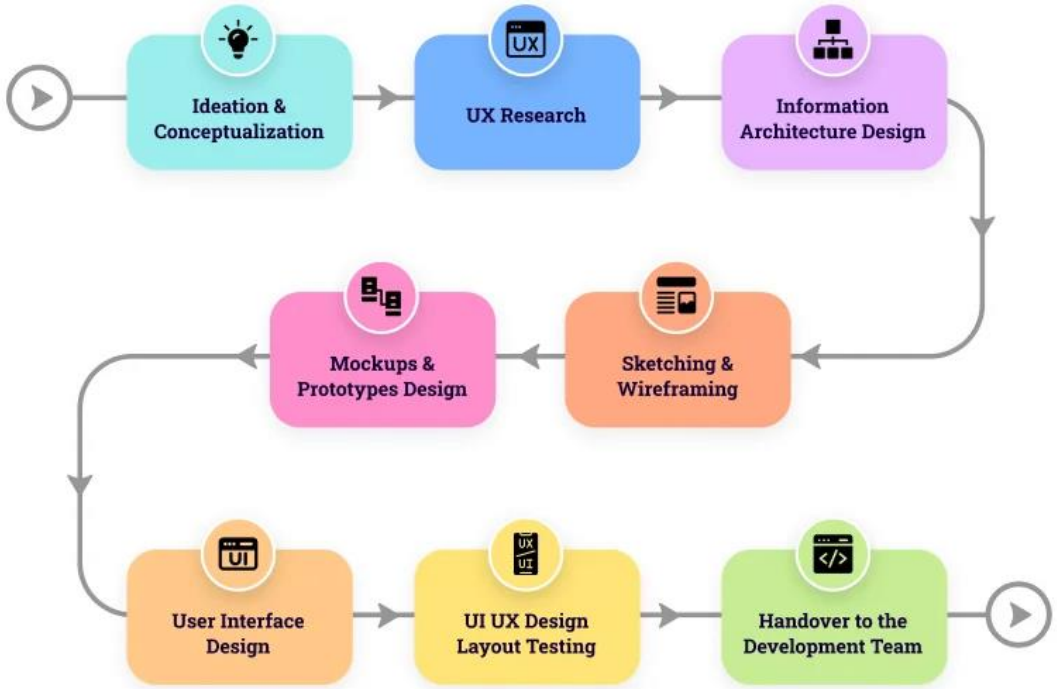
- Great apps balance both, aesthetic and usability
- Example: a beautiful app (UI) can still be frustrating (bad UX)

UX

- “User Experience” = how the app feels to use
- Logical flow between screens
- Clear goals and feedback (knowing what’s happening)
- Emotional connection and comfort
- Accessibility and inclusivity for all users

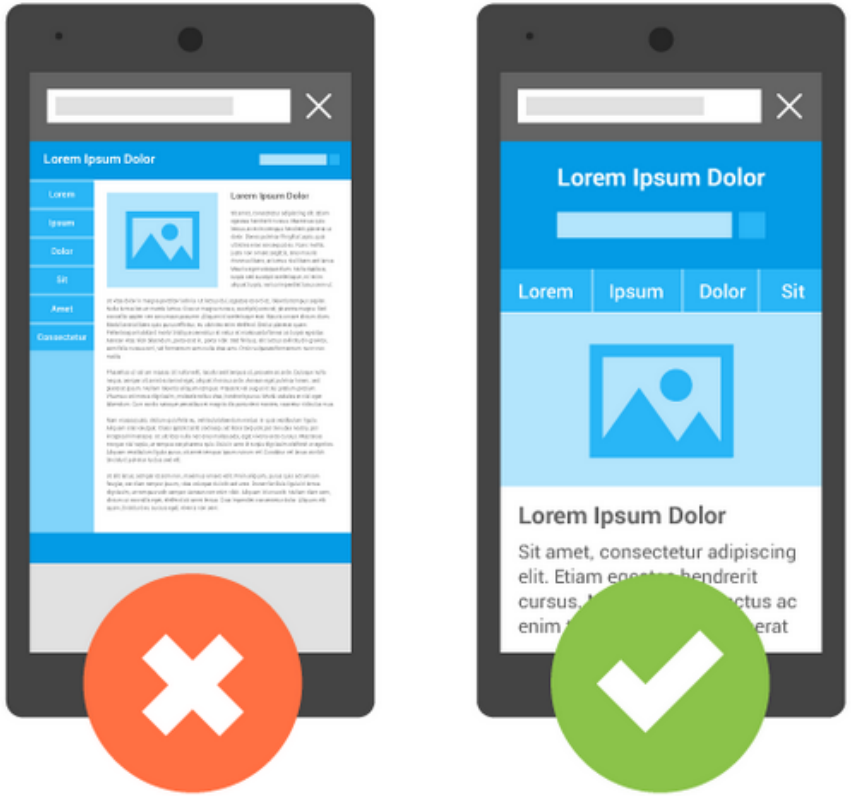
UX VS UI : What's the difference ?

UX (Experience)	UI (Interface)
App structure and logic	Colors and shapes
User journey	Buttons and icons
Ease of navigation	Visual aesthetics
Feels intuitive	Feels beautiful

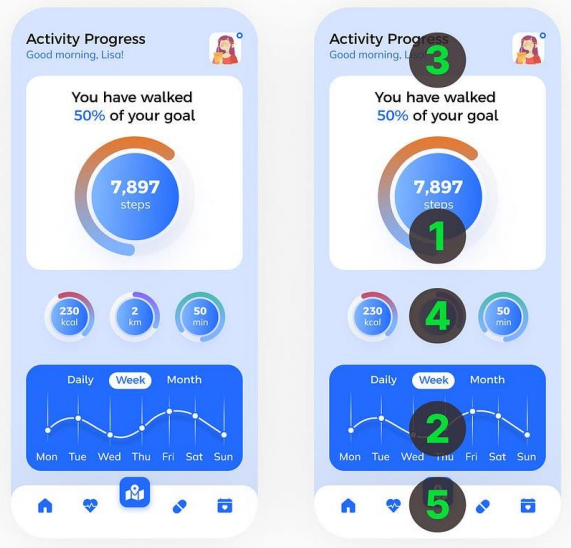


Mistakes to Avoid

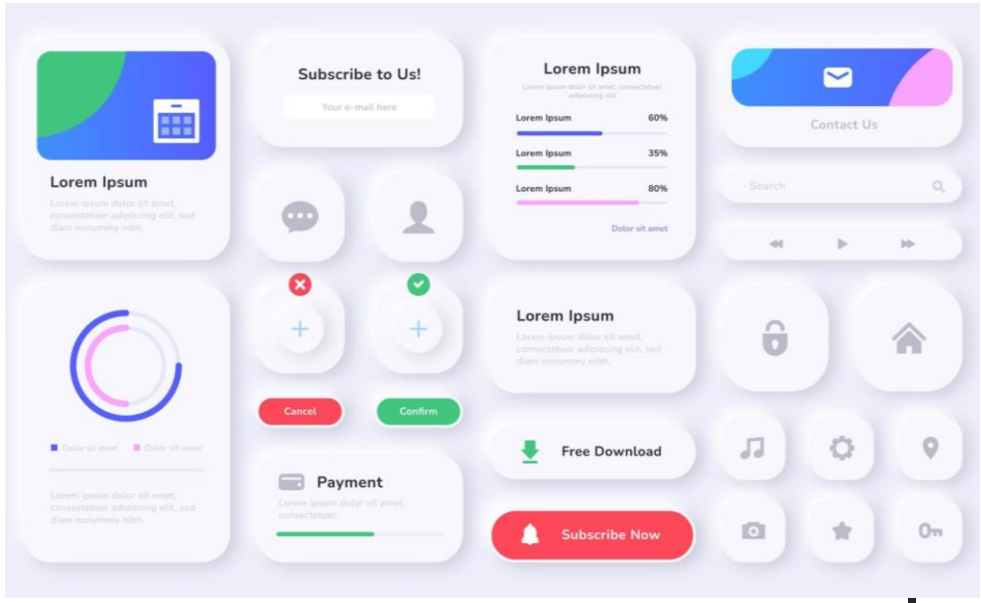
- Small or too-close buttons
- Too many buttons / cluttered layout
- Poor contrast or small text
- Hidden navigation or confusing icons
- No accessibility attributes or feedback
- Ignoring tablet and foldable formats



Design rules that guide good mobile UX/UI



- Simplicity and minimalism
- Consistency across screens
- Visual hierarchy (important elements first)
- Feedback on user actions (button animation)
- Accessibility for all users



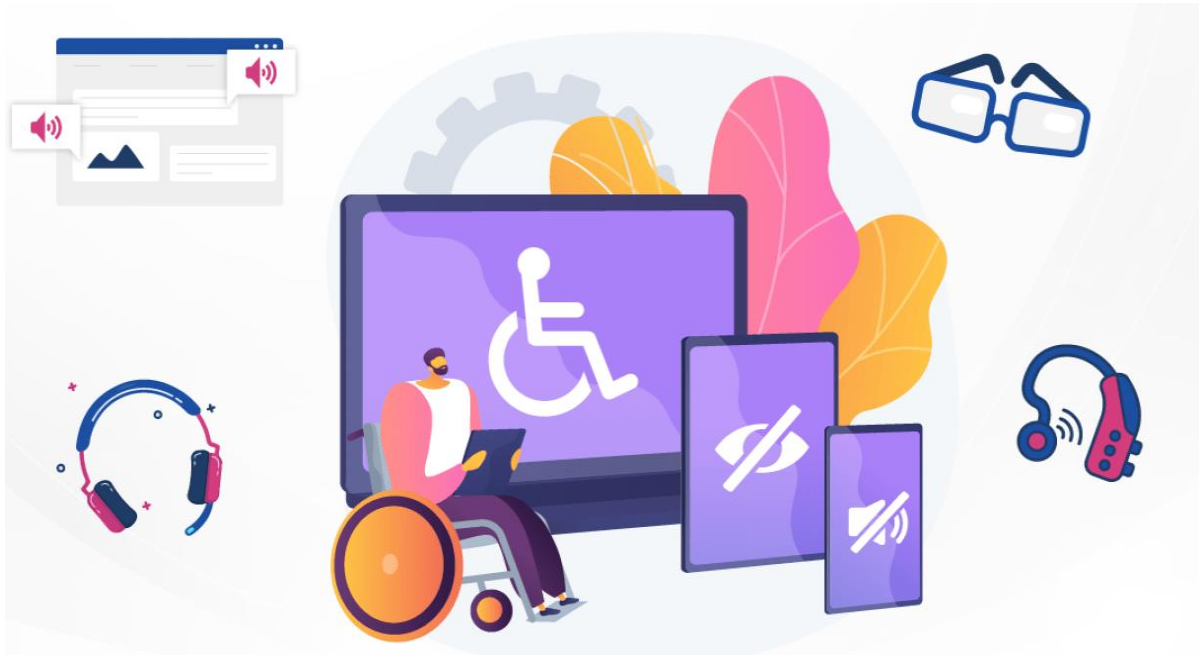
Accessibility = Inclusive Design

1 in 6 people lives with a disability (vision, motor, cognitive, hearing).

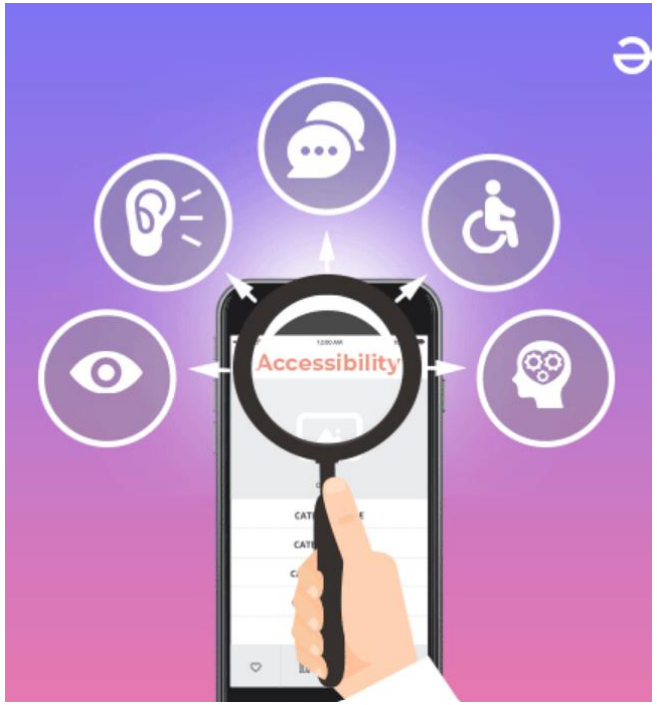
Accessibility improves **usability for everyone**, not just disabled users.

Accessible apps have **higher retention and fewer errors.**

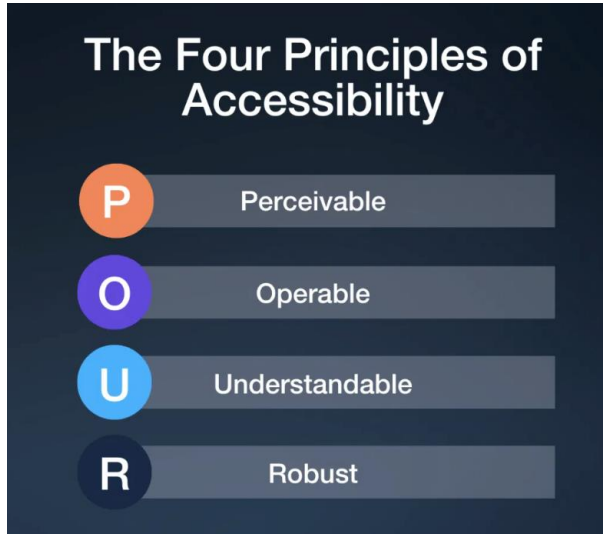
Required by **international guidelines** (WCAG, ADA...).



Design for everyone

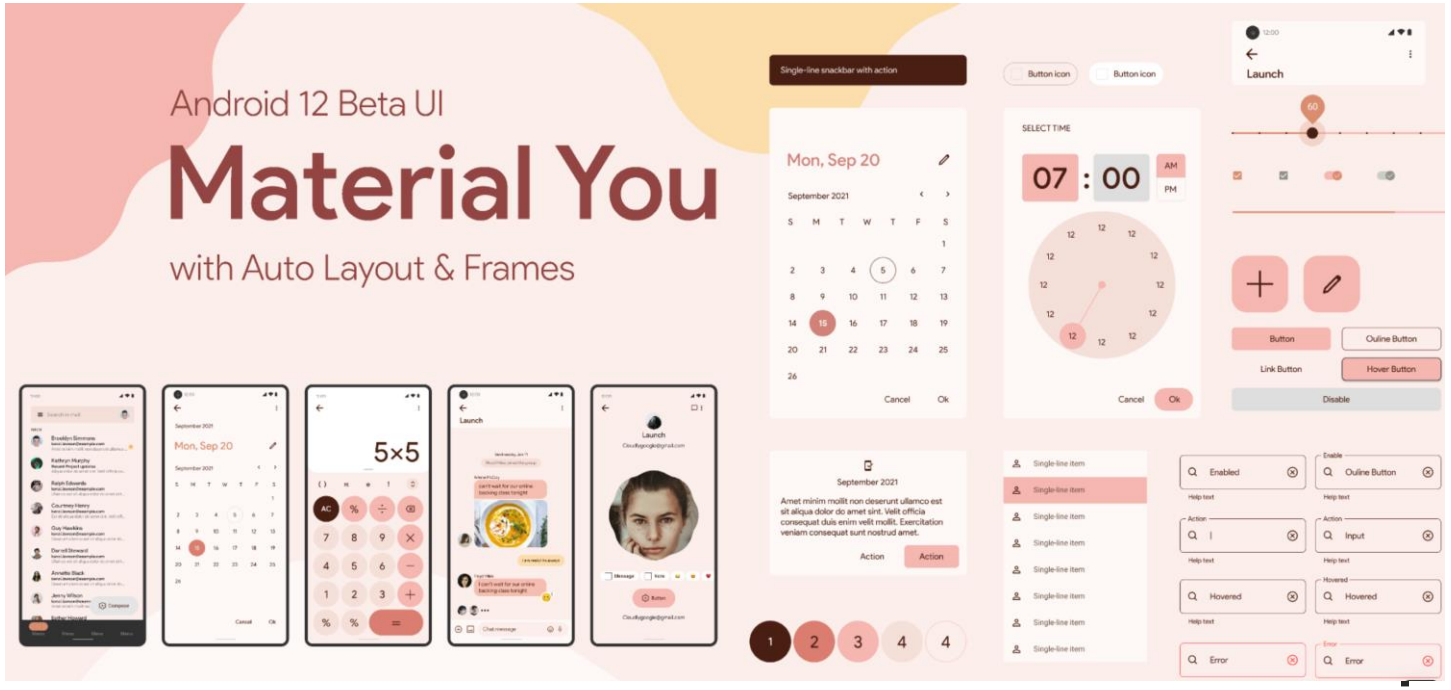


- Design usable **by everyone**
- Support for **screen readers** and voice commands
- **High contrast & readable text**
- **Large touch areas** for easy interaction
- **Simple layouts & clear navigation**



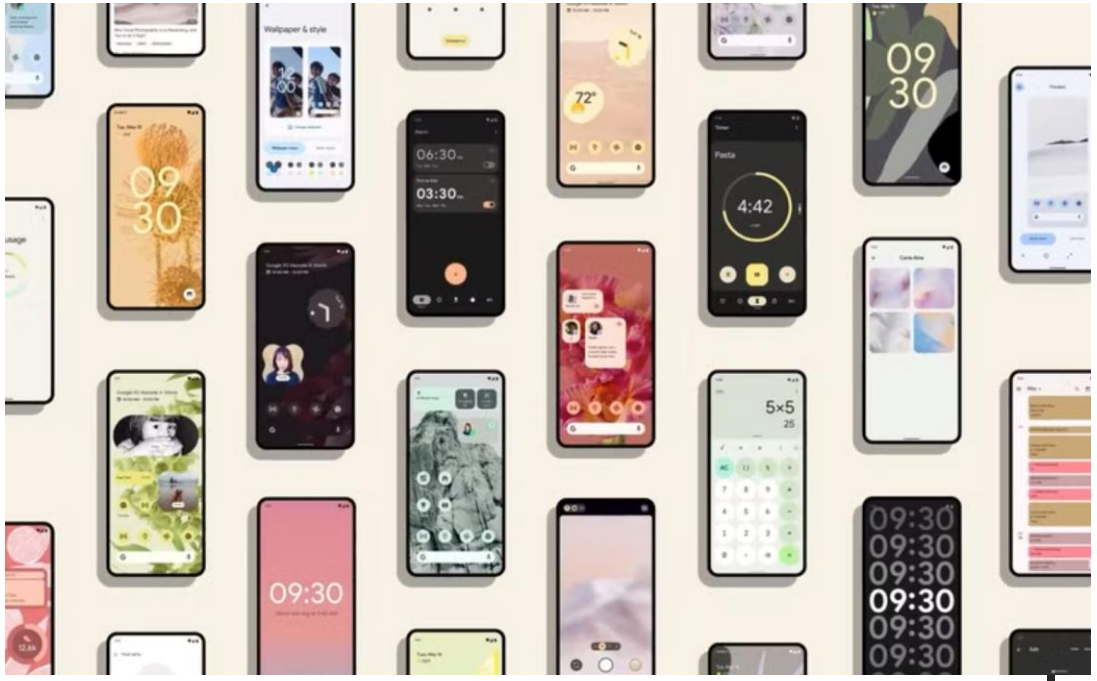
Material You - Google's Modern Design System

- **Dynamic system** that adapts the UI to the user's device colors & preferences
- **Focus on personalization**, expressive design, and accessibility
- **Follows clear rules for spacing**, motion, harmony, and readability
- **Official design language** for Android 12+



Core Principles of Material You

- Dynamic color
- Rounded components that reflect user personality
- Meaningful motion
- UI colors adapt to the system wallpaper
- Consistent spacing & hierarchy
- Smooth transitions to support comprehension
- Shape flexibility
- Clear structure that guides the eyes naturally
- Contrast, readable font scaling, large touch targets



How to enable Material You in an Android app

- Automatically adapts colors to the system palette (Material You)
- Ensures proper contrast and accessibility
- Removes the need for manually defining large color sets

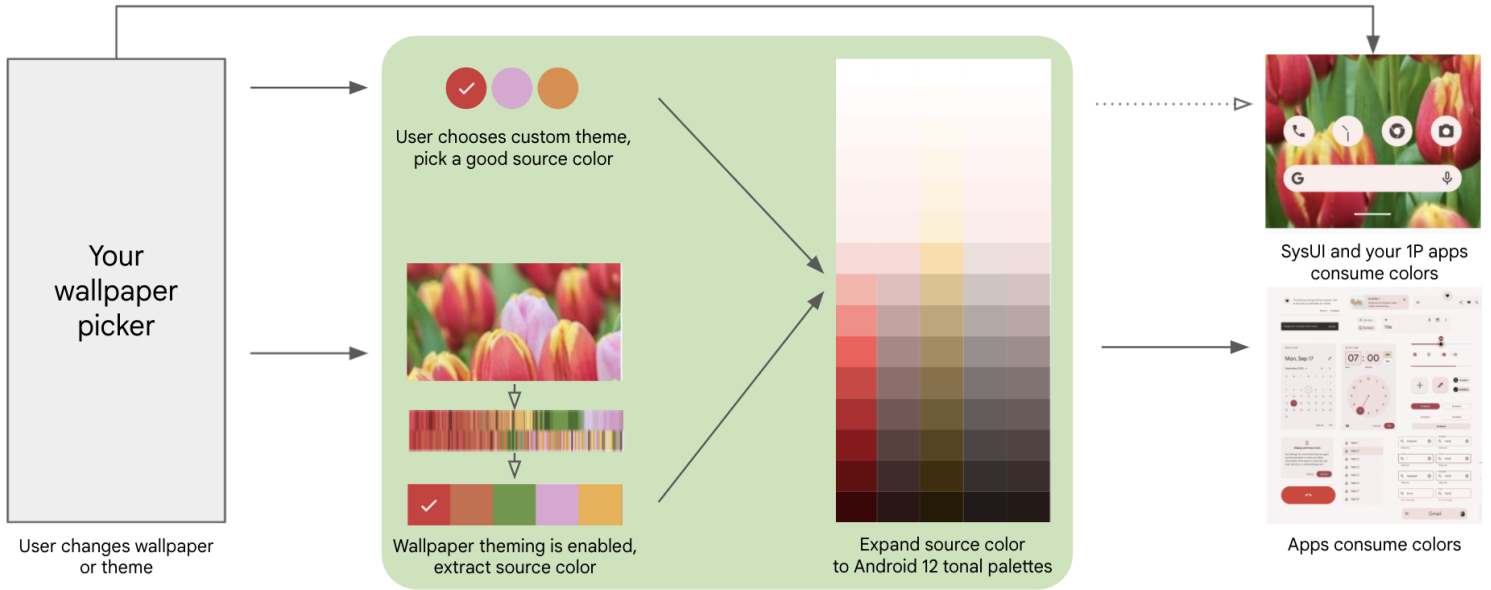
```
@Composable
fun MyTheme(
    darkTheme: Boolean = isSystemInDarkTheme(),
    dynamicColor: Boolean = true,
    content: @Composable () -> Unit
) {
    val context = LocalContext.current

    val colorScheme =
        if (dynamicColor && Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
            if (darkTheme) dynamicDarkColorScheme(context)
            else dynamicLightColorScheme(context)
        } else {
            if (darkTheme) darkColorScheme()
            else lightColorScheme()
        }

    MaterialTheme(
        colorScheme = colorScheme,
        typography = Typography(),
        content = content
    )
}
```

Material You components in practice

- Buttons adapt **shape + elevation + dynamic color**
- Navigation bars **follow system theme**
- Cards use **rounded corners & content hierarchy**
- Typography **automatically** scales with user settings
- Icons inherit colorScheme highlights



All partners use Android 12 implementation for this part of the flow

Démonstration